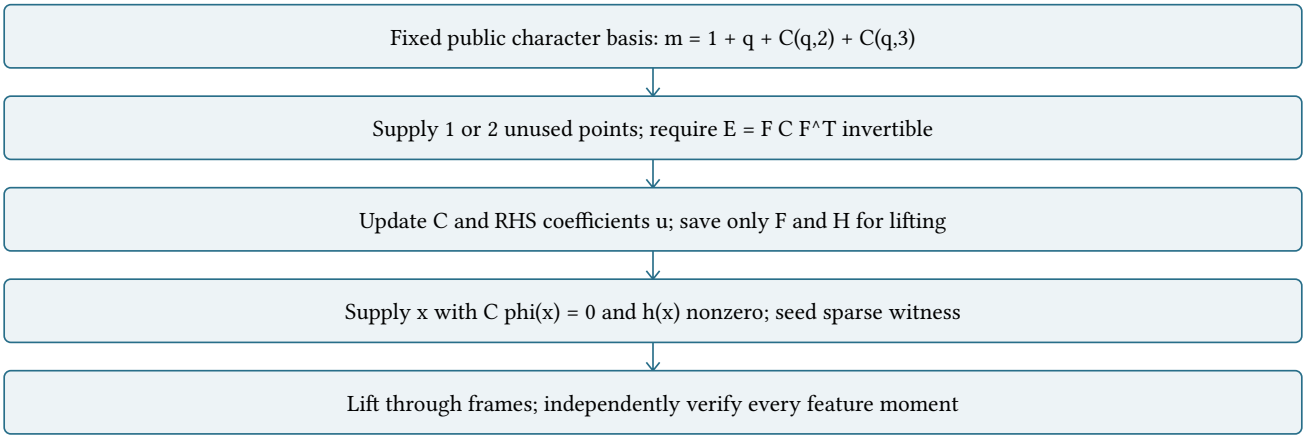
Feature-coefficient Schur and sparse witnesses

2026-11-03 · Derived construction / static audit · Numerical trials: zero

Closed representation. The preserved public character features are exactly all subsets of a q-window of size at most three. Represent the Gram residual by C and its right-hand side by h(x) = x − phi(x)^T u. No r-row array is allocated.



Exact update. For supplied pivot rows F, compute W = C F^T, E = F W and H = W E^(-1). Then C_new = C − H W^T and u_new = u + H h_I. Reject singular blocks. The basis stays fixed; state is immutable. This is the established Schur identity specialized to the preserved feature family.

Sparse certificate. A supplied terminal point must satisfy C phi(x)=0 and h(x) nonzero. Normalize its coefficient by 1/h(x). Each frame lifts a tail moment mu using y_I = −H^T mu and mu_old = mu + F^T y_I. At the root, the invertible diagonal C implies zero feature moment. A separate checker recomputes all feature moments and the pairing sum y_x x = 1 from support. Internal provenance tokens are not a substitute for this final check.

Derived quantity	Schedule / storage
State coefficients	m squared + m field entries
Frame coefficients	2mk field entries, k = 1 or 2
Pivot products	2km squared + 2k squared m + 2km
Lift products	2mk
Accepted pivot coordinates	At most min(m,r)
Certificate support	At most m+1

Pivot counts exclude feature evaluations and small-block inversion. One point evaluation uses qL+J products, with L the binary-power count for (r−1)/2 and J the total label incidence. Details and additions are in DERIVATION.md. Native validation, copies, allocations, failed proposals and final checking are separate. Storage is O(q^6), and need not be smaller than r.

Next dependency. Select useful public pivot/terminal points without a field sweep; analyze when the residual affine functional lies in the feature span. The current routines accept supplied points. A feature certificate still needs its explicit target-extraction connection in a complete driver.

Evidence. Source bytes, pattern/power AST and syntax checked statically; no numerical, corruption, heap or performance trial. New trial execution requires fresh admission and frozen controls. Earlier cohorts are unchanged.

Sources. Preserved direct-cyclic-companion/generators.py; Higham (2023), “What Is the Schur Complement of a Matrix?”, elimination and block inverses. No real positive-definiteness assumptions are used in this finite-field work.